

HATTULA LEPAA AWS, Finland

WMO: 027040

Lat: 61.115N

Lon: 24.324E

Elev: 285

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.5	(c) -4.1	(d) -15.0	(e) 2.5	(f) -9.9	(g) -8.2	(h) 3.6	(i) -3.7	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	81.9	65.9	78.3	64.3	74.8	62.7	68.9	77.0	66.6	74.9	64.5	72.3	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	97.2	72.6	63.6	88.9	70.2	61.3	81.9	67.6	33.2	77.5	31.4	75.0	29.7	72.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -12.2	(f) 85.1	(g) 10.7	(h) 4.1	(i) -19.8	(j) 88.0	(k) -26.1	(l) 90.4	(m) -32.1	(n) 92.7	(o) -39.8	(p) 95.7		
(4) N/A	(5) N/A	(6) N/A	(e) -12.5	(f) 70.8	(g) 10.6	(h) 3.8	(i) -20.2	(j) 73.6	(k) -26.4	(l) 75.8	(m) -32.4	(n) 77.9	(o) -40.1	(p) 80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	20.2	22.5	28.4	39.4	51.5	58.1	63.8	60.6	52.2	40.6	34.2	26.7		
	DBStd	16.75	11.08	11.73	8.45	6.49	7.55	6.17	5.83	5.04	5.71	6.25	7.38	10.20		
	HDD50	4300	923	771	668	323	73	7	0	0	43	296	475	721		
	HDD65	8623	1388	1191	1133	767	422	221	96	154	384	756	925	1186		
	CDD50	1245	0	0	0	6	119	251	427	328	110	4	0	0		
	CDD65	92	0	0	0	0	3	14	58	17	0	0	0	0		
	CDH74	909	0	0	0	0	73	183	522	131	0	0	0	0		
	CDH80	179	0	0	0	0	6	30	122	21	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	23.2	1.9	1.3	1.2	1.2	1.5	2.5	2.8	2.7	1.9	2.5	2.0	1.8		
(15) Precipitation	PrecMax	26.7	3.1	2.9	2.2	2.8	3.1	4.4	6.5	5.3	5.5	6.3	4.2	3.8		
	PrecMin	17.4	0.3	0.1	0.2	0.1	0.4	1.1	0.2	0.5	0.4	0.7	0.2	0.2		
	PrecStd	2.4	0.7	0.8	0.5	0.7	0.8	0.9	1.7	1.3	1.3	1.2	1.0	0.8		
	PrecStd	2.4	0.7	0.8	0.5	0.7	0.8	0.9	1.7	1.3	1.3	1.2	1.0	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.6	41.8	50.1	68.6	80.4	83.8	87.3	83.1	71.3	57.2	50.2	46.8		
		MCWB	37.4	37.3	40.9	52.0	59.9	65.5	68.1	66.8	62.2	53.6	48.4	45.7		
	2%	DB	36.7	38.3	44.8	61.3	75.5	79.2	83.6	77.3	66.7	53.9	48.0	41.8		
		MCWB	35.8	36.3	38.8	47.6	58.6	63.2	67.8	64.7	59.4	51.4	47.0	40.6		
	5%	DB	34.8	36.5	41.2	56.5	71.4	75.1	79.9	74.0	63.7	52.2	45.2	39.3		
		MCWB	34.1	35.2	36.7	45.8	57.3	61.6	66.4	62.9	57.5	50.3	44.2	38.3		
	10%	DB	33.4	34.8	38.4	51.5	66.9	71.1	76.1	70.7	61.2	50.1	43.1	37.3		
		MCWB	32.7	33.9	35.2	42.7	54.2	59.5	65.1	61.1	56.9	48.2	42.3	36.3		
	0.4%	WB	37.9	38.2	42.5	53.3	64.3	68.4	72.4	70.4	63.1	54.7	49.1	45.9		
		MCDB	38.3	39.8	47.6	64.4	74.1	77.3	81.5	77.6	68.7	56.4	49.9	46.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	36.7	39.5	49.8	61.2	65.5	70.1	66.6	60.9	52.4	46.7	40.7		
		MCDB	36.6	38.0	43.4	60.4	71.9	75.9	78.8	73.4	65.2	53.4	47.9	41.5		
	5%	WB	34.2	35.4	37.5	46.5	58.7	62.9	68.1	64.5	59.1	50.7	44.3	38.4		
		MCDB	34.6	36.5	40.3	54.2	67.7	72.2	76.5	71.0	62.3	51.8	45.1	39.1		
	10%	WB	32.8	34.0	35.8	43.8	55.6	60.7	66.0	62.7	57.3	48.3	42.4	36.4		
		MCDB	33.3	34.8	38.1	50.2	65.6	69.3	74.3	68.5	60.2	49.9	43.0	37.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	10.3	14.0	17.2	19.7	18.0	17.9	16.9	13.5	10.1	6.5	7.5		
		MCDBR	6.0	8.5	15.1	26.7	27.7	25.3	23.3	22.3	17.6	11.6	7.7	8.0		
		MCWBR	5.9	7.2	10.0	14.6	13.2	11.9	10.0	11.0	11.0	9.1	7.5	7.8		
		MCDBR	5.9	8.0	13.7	23.5	23.4	22.3	20.7	19.0	15.6	10.2	7.6	8.1		
	5% WB	MCWBR	5.8	7.0	9.7	13.7	12.5	11.5	10.0	10.3	10.5	8.9	7.5	7.9		
		MCWBR	5.8	7.0	9.7	13.7	12.5	11.5	10.0	10.3	10.5	8.9	7.5	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.297	0.325	0.356	0.353	0.350	0.373	0.362	0.336	0.323	0.281	0.368		
	taud		2.091	2.165	2.246	2.375	2.440	2.472	2.424	2.449	2.515	2.408	2.192	2.616		
	Ebn at Noon		146	212	250	266	276	279	268	262	250	207	136	99		
	Edh at Noon		13	22	29	31	32	31	32	29	23	18	11	8		
(44) All-Sky Solar Radiation	RadAvg		74	253	721	1187	1604	1681	1618	1255	747	320	87	41		
	RadStd		14	56	91	125	175	132	159	140	98	56	16	5		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-252	N/A	N/A	N/A

Nomenclature: See separate page